

Abdul Rahaman S

Machine Learning Engineer | Data Scientist

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PROFILE

Machine Learning Engineer and Data Scientist with strong expertise in **machine learning, deep learning, NLP, and model evaluation**, combined with hands-on experience supporting and observing **production ML systems** in a global B2B environment. Proven ability to design **end-to-end ML solutions**, build explainable models, and translate real-world system behavior into robust learning pipelines. Actively transitioning from AI-adjacent production support into full-time ML engineering.

SKILLS

Languages

Python, SQL

Deep Learning & NLP:

TensorFlow, Keras, Transformers, LLM, RAG

MLOps & Deployment

MLflow, DVC, Docker, GitHub Actions, FastAPI, CI/CD

Cloud & Systems

AWS (S3, EC2, SageMaker), Azure (AZ-900, AZ-104), Git, PostgreSQL

Machine Learning & Data Science:

Scikit-learn, XGBoost, Pandas, NumPy, Model Evaluation, Feature Engineering

PROJECTS

Fraud Detection System – End-to-End MLOps Project 🔗

Technologies: Python, XGBoost, Scikit-learn, FastAPI, MLflow, Docker, GitHub Actions, SHAP

- Trained an **XGBoost-based fraud classifier** on highly imbalanced transaction data (~0.17% fraud rate), improving fraud recall compared to baseline models.
- Applied **business-driven threshold tuning**, achieving a stronger recall–precision balance than default probability thresholds.
- Achieved **high ROC-AUC (high 0.9s)** while prioritizing recall for fraud detection use cases.
- Integrated **SHAP explainability** to provide feature-level transparency for both global trends and individual predictions.
- Deployed a **FastAPI inference service**, containerized using Docker, with automated CI checks and experiment tracking via MLflow.

RAG Chatbot (NLP / LLM Project) 🔗

Technologies: python, FastAPI, Ollama, Sentence-transformers, chromaDB

- Developed a **fully offline RAG system** to answer questions grounded in PDFs and lecture videos.
- Implemented **hybrid retrieval** using vector similarity (ChromaDB) and keyword-based scoring.
- Built an idempotent ingestion pipeline with PDF parsing, Whisper-based transcription, chunking, and embeddings.
- Integrated **local LLM inference** (Ollama) with strict context enforcement to reduce hallucinations.

- Improved answer relevance and reduced hallucinations through hybrid retrieval and strict context-only generation.

PROFESSIONAL EXPERIENCE

Ciklum (Pixellot AI Sports Production) Mar 2024 – present | Chennai
Support Engineer

- Supported a **large-scale sports production platform** used by globally distributed clients in a B2B environment.
- Monitored system and stream health using **Grafana dashboards**, identifying anomalies early to **prevent downstream failures**.
- Investigated application and system logs to diagnose **pipeline failures, data inconsistencies, and configuration issues**.
- Conducted acceptance testing for new installations to ensure **system stability prior to client handover**.
- Escalated recurring and critical issues to R&D teams with **clear reproduction steps, logs, and environment context**, contributing to long-term system reliability.
- Exposure to production systems and algorithm behavior directly motivated deeper specialization in **machine learning and applied AI development**.

Infosys Oct 2020 – Mar 2024 | Chennai
System Engineer

- Resolved Windows OS and enterprise application incidents within SLA using ServiceNow.
- Performed patching, vulnerability remediation, and system administration across servers and workstations.
- Supported Office 365 services including Outlook, Teams, OneDrive, and SharePoint.
- Contributed to operational improvements through proactive monitoring and issue prevention.

CERTIFICATES

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| • AWS Certified Machine Learning Engineer – Associate (MLA-C01) | • Microsoft Certified: Azure Administrator Associate (AZ-104) | • Microsoft Certified: Azure Fundamentals (AZ-900) |
| • ITIL V4 Foundation – Service Management Best Practices | • Business Communication Excellence – Level 5 | |

EDUCATION

University of madras 2017 – 2020 | Chennai, India
Bachelor of computer applications

Dhanalakshmi School 2017 | Chennai
Higher Secondary